The Evolution of Redox Theory — NCERT Three-Tier Conceptual Hierarchy

Redox processes govern chemistry from metabolic respiration and industrial metallurgy to battery electrochemistry. NCERT develops the concept across three historical tiers: Classical (Oxygen/Hydrogen) → Electronic (Electron transfer) → Oxidation Number (Formal bookkeeping).

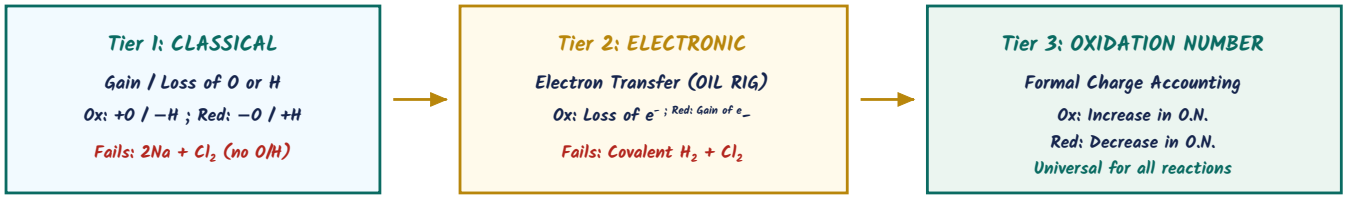


Fig. 1.1 — The conceptual evolution of Redox from Classical to Electronic to Oxidation Number bookkeeping.

1. Historical Breakdown of the Three Tiers

Tier	Core Definition of Oxidation	Core Definition of Reduction	Where the Model Breaks Down
Tier 1: Classical (7.1)	**Addition of Oxygen** or electronegative element; **Removal of Hydrogen** or electropositive element	**Removal of Oxygen** or electronegative element; **Addition of Hydrogen** or electropositive element	Fails in reactions with no O or H (e.g. $2Na + Cl_2 \rightarrow 2NaCl$). Misleads in non-redox oxygen transfers ($CaO + SO_3 \rightarrow CaSO_4$).
Tier 2: Electronic (7.2)	**Loss of electrons** (OIL: Oxidation Is Loss). Species becomes more positive / less negative.	**Gain of electrons** (RIG: Reduction Is Gain). Species becomes less positive / more negative.	Fails in covalent reactions where electrons are merely shifted rather than completely transferred (e.g. $H_2 + Cl_2 \rightarrow 2HCl$).
Tier 3: Oxidation Number (7.3)	**Increase in Oxidation Number (O.N.)** of an element in the given species.	**Decrease in Oxidation Number (O.N.)** of an element in the given species.	Universal operational tool! However, O.N. is a formal book-keeping device; it does NOT imply real physical ionic charges.

Table 1.1 — Comparative operational definitions across the three tiers of redox theory.

2. Fundamental Terminology in Redox

- **Oxidising Agent (Oxidant):** The reactant that supplies electronegative atoms, removes electrons, or undergoes a **decrease in oxidation number**. It oxidises others and is **itself reduced**.
- **Reducing Agent (Reductant):** The reactant that supplies electropositive atoms, donates electrons, or undergoes an **increase in oxidation number**. It reduces others and is **itself oxidised**.
- **Redox Couple (7.4):** A pair consisting of the oxidised and reduced forms of the same substance taking part in an oxidation or reduction half-reaction, denoted as **Ox / Red** (e.g. Cu^{2+} / Cu, Fe^{3+} / Fe^{2+}).

Oxidation Number Rules — The Rigorous Book-Keeping Protocol (NCERT 7.3)

Oxidation number denotes the residual charge which an atom appears to have when all other atoms from the molecule are removed as ions by counting shared electrons towards the more electronegative partner.

1. The Master Axioms for Assigning Oxidation State

Element / Environment	Assigned Oxidation State	Exceptions & Severe JEE Traps
Elemental / Free State	**Strictly 0**	O_2, O_3, P_4, S_8, C_{60}, Fe, Cl_2, H_2 all have O.S. = 0.
Fluorine (F)	**Always −1** in ALL compounds	**NO EXCEPTIONS!** F is the most electronegative element (Pauling 4.0). In HOF, F is −1, O is 0, H is +1.
Alkali Metals (Group 1: Li, Na, K, Rb, Cs)	**Always +1** in all compounds	**NO EXCEPTIONS!** Never exhibits +2 or 0 in compounds.
Alkaline Earths (Group 2: Be, Mg, Ca, Sr, Ba)	**Always +2** in all compounds	**NO EXCEPTIONS!** Only two valence s-electrons.
Hydrogen (H)	**+1** with non-metals	**−1 in metal hydrides** (ionic hydrides: NaH, CaH_2, $LiAlH_4$, $NaBH_4$, where metal is less EN).
Oxygen (O)	**Normally −2** (Oxides: H_2O, CO_2, MgO)	· **−1 in Peroxides:** H_2O_2, Na_2O_2, BaO_2 · **−½ in Superoxides:** KO_2, RbO_2, CsO_2 · **+2 in OF_2** (F is more EN than O) · **+1 in O_2F_2** (dioxygen difluoride) · **0 in O_2 and O_3; +½ in O_2^+.**
Halogens (Cl, Br, I)	**Normally −1** (Halides)	**Positive O.S. (+1 to +7)** when bonded to more EN atoms (O and F)! E.g. ClO_4^- (+7), Cl_2O_7 (+7), IF_7 (+7), ICl (+1 for I).

Table 1.2 — Universal hierarchical rules for assigning oxidation states.

2. Algebraic Sum Rule for Neutral and Polyatomic Species

KEY RESULT

· In a **neutral molecule**, the algebraic sum of oxidation numbers of all atoms is **identically 0**:
For A_xB_y: **$x \cdot O.S.(A) + y \cdot O.S.(B) = 0$**

· In a **polyatomic ion**, the algebraic sum equals the **net electrical charge** of the ion:
For $[SO_4]^{2-}$: $O.S.(S) + 4(-2) = -2 \Rightarrow$ **$O.S.(S) = +6$**

· **Periodic Range Limit:** Maximum positive O.S. of a representative element **CANNOT exceed its valence shell group number** (e.g. S cannot exceed +6, P cannot exceed +5, Cr cannot exceed +6). If naive algebra yields > Group Number, look for a **peroxide bridge**!

Peroxide Traps in Oxidation States — When Naive Algebra Fails

Blind algebraic calculation often produces impossible oxidation states that violate the octet rule or exceed the total valence electrons of the atom. Whenever an atom appears to exceed its maximum valence shell group number, draw the Lewis structure to locate peroxo (–O–O–) linkages.

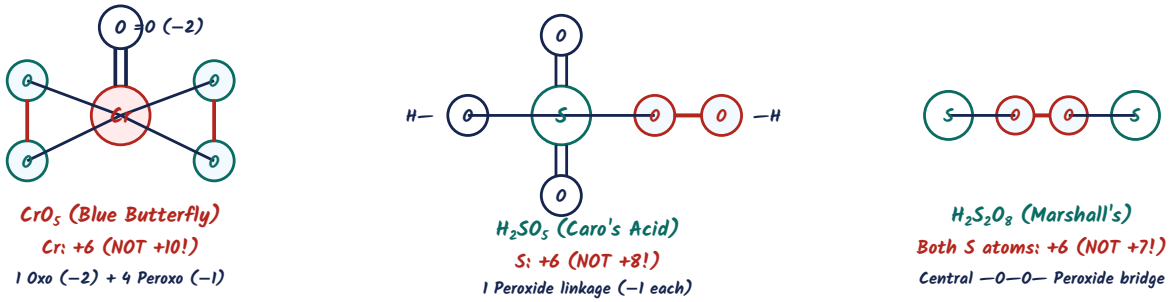


Fig. 1.2 — Structures of classic peroxide exceptions: CrOS (Butterfly), H_2SO_5 (Caro's), and $H_2S_2O_8$ (Marshall's).

1. High-Yield Peroxide Traps Comparison

Compound	Naive Formula O.S.	True Structural O.S.	Structural Rationale & Peroxide Bonds
Chromium Pentoxide (CrO_5)	$Cr + 5(-2) = 0 \Rightarrow$ **Cr = +10 (IMPOSSIBLE!)**	**Cr = +6**	Butterfly structure: **1 oxo oxygen (=O, –2)** and **4 peroxo oxygens (–O–O–, –1 each)**. Cr has 6 valence electrons!
Caro's Acid (H_2SO_5)	$2(+1) + S + 5(-2) = 0 \Rightarrow$ **S = +8 (IMPOSSIBLE!)**	**S = +6**	Peroxomonosulphuric acid: contains one **–O–O–H** peroxide group (–1 each) and three normal oxygens (–2 each). S is max +6.
Marshall's Acid ($H_2S_2O_8$)	$2(+1) + 2S + 8(-2) = 0 \Rightarrow$ **S = +7 (IMPOSSIBLE!)**	**Both S = +6**	Peroxodisulphuric acid: two $-SO_3H$ units linked by a central **–O–O– peroxide bridge** (–1 each).
Pyrosulphuric Acid ($H_2S_2O_7$)	$2(+1) + 2S + 7(-2) = 0 \Rightarrow$ **S = +6**	**Both S = +6**	Oleum: contains an **S—O—S bridge**; NO peroxide link!

Table 1.3 — Peroxide species where naive algebraic sum violates maximum oxidation states.

KEY RESULT

Peroxide Detection Rule:

If calculated O.S. exceeds maximum group valence (e.g. S in H_2SO_5 = +8, Cr in CrO_5 = +10):

- The actual oxidation state of the central atom is equal to its **maximum group valence** (S = +6, Cr = +6)!
- The excess positive charge is accounted for by **peroxide oxygen atoms having O.S. = –1** instead of –2.

Fractional Oxidation Numbers & Mixed Oxides — Atomic Reality vs Averages

Fractional oxidation states (e.g. +8/3, +4/3, +2.5) are purely **mathematical averages** resulting from non-equivalent atoms of the same element in different oxidation states within the same chemical formula unit (NCERT 7.3.2).



Fig. 1.3 — Fractional oxidation state structures: Carbon Suboxide (C_3O_2) and Tribromooctaoxide (Br_3O_8).

1. Classic Mixed-Valence Solids

Formula Unit	Average Formula O.S.	True Stoichiometric Composition	Individual Atomic Oxidation States
Magnetite (Fe_3O_4)	**+8/3** (+2.67)	Mixed stoichiometric spinel: **$FeO \cdot Fe_2O_3$**	One **Fe^{2+} (+2)** and two **Fe^{3+} (+3)** ions.
Red Lead (Pb_3O_4 / Sindoor)	**+8/3** (+2.67)	Mixed oxide: **$2\,PbO \cdot PbO_2$**	Two **Pb^{2+} (+2)** and one **Pb^{4+} (+4)** ions.
Hausmannite (Mn_3O_4)	**+8/3** (+2.67)	Mixed oxide: **$2\,MnO \cdot MnO_2$**	Two **Mn^{2+} (+2)** and one **Mn^{4+} (+4)** ions.
Sodium Tetrathionate ($Na_2S_4O_6$)	**+2.5** (+5/2)	$^{-}O_3S—S—S—SO_3^{-}$	Two central bridging S atoms = **0**; two terminal S atoms = **+5**.
Bleaching Powder ($CaOCl_2$)	**0**	Mixed salt: **$Ca^{2+}\,(OCl)^{-}\,(Cl)^{-}$**	One hypochlorite Cl = **+1**; one chloride Cl = **−1**!
Ammonium Nitrate (NH_4NO_3)	**+1**	Two discrete ions: **NH_4^{+} and NO_3^{-}**	Nitrogen in NH_4^{+} = **−3**; Nitrogen in NO_3^{-} = **+5**!

Table 1.4 — Deconvolution of fractional and ambiguous average oxidation numbers.

Charge Concepts in Chemistry — Dissecting O.N., Formal Charge & Partial Charge

A common source of confusion in JEE Advanced is equating oxidation state with formal charge or partial real charge. They represent three distinct theoretical frameworks calculated under entirely different assumptions.

1. Conceptual Comparison of the Three Charges

Feature	Oxidation Number (O.N.)	Formal Charge (F.C.)	Real Partial Charge (δ)
Underlying Assumption	**100% Ionic Limit:** All bonding electrons are assigned entirely to the more electronegative atom.	**100% Covalent Limit:** All bonding electrons are shared equally (50:50) regardless of EN difference.	**Actual Physical Charge:** Electron density calculated from dipole moments or quantum wavefunction.
Calculation Formula	$O.N. = V - N_{lone} - N_{assigned}$	$F.C. = V - L - \frac{1}{2} S$	$\delta = q / e$ (fractional continuum between −1 and +1)
Sensitivity to Electronegativity	**Extremely sensitive** (step-function: winner takes all)	**Completely blind** to electronegativity differences	**Gradual function** of electronegativity difference
Example: HCl	H = **+1**, Cl = **−1**	H = **0**, Cl = **0**	H = **+0.18**, Cl = **−0.18**
Example: CO	C = **+2**, O = **−2**	C = **−1**, O = **+1** (due to dative bond!)	C = **−0.11**, O = **+0.11** (nearly non-polar, μ = 0.11 D)

Table 1.5 — Rigorous contrast between Oxidation Number, Formal Charge, and Real Charge.

2. Rules for Assigning O.N. from Lewis Structures

KEY RESULT

When dealing with organic molecules or coordination complexes, assign O.N. bond-by-bond:

- **Homonuclear Bond (A—A):** Shared pair divided equally → contribution to both atoms = **0**.
- **Heteronuclear Single Bond (A—B where B is more EN):** Both electrons go to B → **A gets +1, B gets −1**.
- **Heteronuclear Double/Triple Bond:** A gets **+2 / +3**, B gets **−2 / −3**.
- **Coordinate Dative Bond (A → B):**
  - If donor A is less EN than B: A gets **+2**, B gets **−2**.
  - If donor A is more EN than B: Electron pair stays with A → contribution to both = **0**!

Types of Redox Reactions (NCERT 7.3.1) — Mechanistic Classification

Redox reactions are classified into four major categories based on the structural rearrangements of reactants and products: Combination, Decomposition, Displacement (Metal & Non-Metal), and Disproportionation.

1. Combination & Decomposition Reactions

Reaction Type	General Form	Key Criteria & Examples	Non-Redox Traps to Watch
Combination (7.3.1 a)	A + B → C	At least one of A or B must be in **elemental form** (O.S. = 0): · $C(0) + O_2(0) \rightarrow C(+4)O_2(-2)$ · $3Mg(0) + N_2(0) \rightarrow Mg_3(+2)N_2(-3)$ · $CH_4(-4) + 2O_2(0) \rightarrow CO_2(+4) + 2H_2O(-2)$	$CaO(+2,-2) + SO_3(+6,-2) \rightarrow CaSO_4(+2,+6,-2)$ is **COMBINATION BUT NOT REDOX**! No change in O.S.!
Decomposition (7.3.1 b)	C → A + B	Breakdown of a compound into two or more components, where at least one product is an **element**: · $2H_2O(+1,-2) \rightarrow 2H_2(0) + O_2(0)$ · $2KClO_3(+1,+5,-2) \rightarrow 2KCl(+1,-1) + 3O_2(0)$	$CaCO_3(+2,+4,-2) \xrightarrow{(\Delta)} CaO(+2,-2) + CO_2(+4,-2)$ is **DECOMPOSITION BUT NOT REDOX**! All O.S. unchanged!

Table 1.6 — Combination and decomposition redox pathways contrasted with non-redox analogs.

2. Displacement Reactions (7.3.1 c)

- **Metal Displacement:** A metal in a compound is displaced by another metal in the uncombined state. The displacing metal must have a **lower standard reduction potential (higher oxidation tendency)**:
  · **$CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s)$** (Zn is oxidised 0 → +2; Cu is reduced +2 → 0).
  · **$TiCl_4(l) + 2Mg(s) \rightarrow Ti(s) + 2MgCl_2(s)$** (Kroll process for titanium extraction).
  · **$Cr_2O_3(s) + 2Al(s) \rightarrow Al_2O_3(s) + 2Cr(s)$** (Aluminothermic Goldschmidt reduction).
- **Non-Metal Displacement (Hydrogen Displacement):**
  · **Active metals (Group 1 & 2: Na, Ca):** Displace H_2 from cold water: **$2Na + 2H_2O \rightarrow 2NaOH + H_2\uparrow$**.
  · **Less active metals (Mg, Fe):** Displace H_2 only from steam: **$Mg + H_2O(g) \rightarrow MgO + H_2\uparrow$**; **$3Fe + 4H_2O(g) \rightarrow Fe_3O_4 + 4H_2\uparrow$**.
  · **Metals above H in ECS (Zn, Fe):** Displace H_2 from dilute non-oxidising acids (HCl, dil H_2SO_4): **$Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$**.
- **Halogen Displacement Activity Order: $F_2 > Cl_2 > Br_2 > I_2$** (decreases down group):
  · $Cl_2(g) + 2KBr(aq) \rightarrow 2KCl(aq) + Br_2(l)$
  · $Cl_2(g) + 2KI(aq) \rightarrow 2KCl(aq) + I_2(s)$
  · F_2 is so ferocious an oxidant that it attacks water directly: **$2F_2 + 2H_2O \rightarrow 4HF + O_2\uparrow$** (cannot displace halogens in aqueous solution!).

Auto-Redox Reactions — Disproportionation and Comproportionation Criteria

In auto-redox reactions, the same element acts simultaneously as the oxidising agent and the reducing agent. This requires the element to exist in an **intermediate oxidation state** capable of moving both up and down.

1. Disproportionation Reactions (7.3.1 d)

KEY RESULT

Disproportionation Rule: A reaction in which an element in one oxidation state is **simultaneously oxidised and reduced**.

· **Condition:** The reacting element must possess at least **three oxidation states**, and must be present in an **intermediate** state.

· Elements in their maximum O.S. (e.g. HNO_3 where N is +5, $KMnO_4$ where Mn is +7, H_2SO_4 where S is +6) **CANNOT disproportionate**!

Reacting Element	Intermediate State	Disproportionation Reaction	Products & Final O.S.
Hydrogen Peroxide (O)	O is **−1**	$2\ H_2O_2(-1) \rightarrow 2\ H_2O(-2) + O_2(0)$	Reduced to **H_2O (−2)**; Oxidised to **O_2 (0)**
Phosphorus (P_4)	P is **0**	$P_4(0) + 3\ OH^- + 3\ H_2O \rightarrow$ **$PH_3(-3)$ + 3 $H_2PO_2^-(+1)$**	Reduced to **PH_3 (−3)**; Oxidised to **Hypophosphite (+1)**
Sulphur (S_8)	S is **0**	$S_8(0) + 12\ OH^- \rightarrow$ **4 $S^{2-}(-2)$ + 2 $S_2O_3^{2-}(+2)$ + 6 H_2O**	Reduced to **S^{2-} (−2)**; Oxidised to **Thiosulphate (+2)**
Chlorine in Cold Alkali	Cl is **0**	$Cl_2(0) + 2\ NaOH(cold/dil) \rightarrow$ **$NaCl(-1)$ + $NaClO(+1)$ + H_2O**	Reduced to **Cl^- (−1)**; Oxidised to **Hypochlorite (+1)**
Chlorine in Hot Alkali	Cl is **0**	$3\ Cl_2(0) + 6\ NaOH(hot/conc) \rightarrow$ **5 $NaCl(-1)$ + $NaClO_3(+5)$ + 3 H_2O**	Reduced to **Cl^- (−1)**; Oxidised to **Chlorate (+5)**
Fluorine (TRAP!)	F is **0**	$2\ F_2 + 2\ OH^- \rightarrow$ **2 F^- + OF_2 + H_2O**	**NOT DISPROPORTIONATION!** F only reduces (0 → −1). O is oxidised (−2 → +2). F has no positive O.S.!

Table 1.7 — Classic disproportionation reactions and the Fluorine non-disproportionation trap.

2. Comproportionation (Synproportionation)

The exact reverse of disproportionation: **two different oxidation states of the same element react to form a single intermediate oxidation state**:

· **$Ag^{2+} + Ag(s) \rightarrow 2\ Ag^+$** (+2 and 0 → +1)

· **$IO_3^-(+5) + 5\ I^-(-1) + 6\ H^+ \rightarrow 3\ I_2(0) + 3\ H_2O$** (+5 and −1 → 0 in iodometric titrations!)

· **$2\ H_2S(-2) + SO_2(+4) \rightarrow 3\ S(0) + 2\ H_2O$** (Claus process for sulphur recovery).

The Oxidation Number Method (NCERT 7.3.2 a) — Stepwise Algorithm

Based on the fundamental principle that the **total increase in oxidation number** of the oxidised species must equal the **total decrease in oxidation number** of the reduced species.

1. Standard 5-Step Procedural Algorithm

- **Step 1 (Identify O.S.):** Write the skeletal equation and assign oxidation states to all atoms to identify which elements are undergoing oxidation and reduction.
- **Step 2 (Equalize redox atoms):** Multiply species containing the changing elements by integer coefficients so that the number of atoms undergoing redox changes is equal on both sides.
- **Step 3 (Calculate ΔO.N. per formula unit):** Determine the net increase in O.N. per formula unit and net decrease in O.N. per formula unit. Multiply the two reactants by suitable cross-coefficients so that **Total Increase in O.N. = Total Decrease in O.N.**
- **Step 4 (Balance charge via H^+ / OH^-):**
  - In **Acidic Medium:** Add H^+ ions to the side that is deficient in positive charge.
  - In **Basic Medium:** Add OH^- ions to the side that is deficient in negative charge.
- **Step 5 (Balance O and H via H_2O):** Count Oxygen atoms on both sides and add the required number of **H_2O molecules** to the oxygen-deficient side. Verify that Hydrogen atoms automatically balance!

2. Worked Exemplar: Cr2O72− + Fe2+ in Acidic Medium

KEY RESULT

Balancing Step-by-Step:

1. Skeletal: **$Cr_2O_7^{2-}(+6) + Fe^{2+}(+2) \rightarrow Cr^{3+}(+3) + Fe^{3+}(+3)$**

2. Equalize Cr: $Cr_2O_7^{2-} \rightarrow 2\ Cr^{3+}$. Change per Cr = 3 e^-; **Total decrease per $Cr_2O_7^{2-}$ = 2 × 3 = 6**

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3. Fe change: $Fe^{2+} \rightarrow Fe^{3+}$. **Total increase per Fe^{2+} = 1.**

4. Cross-multiply: **1 $Cr_2O_7^{2-}$ + 6 Fe^{2+} → 2 Cr^{3+} + 6 Fe^{3+}**

5. Balance charge: Left has (−2 + 6(+2) = +10); Right has (2(+3) + 6(+3) = +24). Need +14 on Left:
$Cr_2O_7^{2-}$ + 6 Fe^{2+} + 14 H^+ → 2 Cr^{3+} + 6 Fe^{3+}

6. Balance O via H_2O: Left has 7 O; add 7 H_2O to right:
$Cr_2O_7^{2-}$ + 6 Fe^{2+} + 14 H^+ → 2 Cr^{3+} + 6 Fe^{3+} + 7 H_2O (Completely Balanced!)

The Half-Reaction Method (NCERT 7.3.2 b) — Rigorous Charge & Electron Tracking

The half-reaction method splits the overall redox reaction into two separate half-reactions: an **Oxidation Half-Reaction (OHR)** and a **Reduction Half-Reaction (RHR)**. Each is balanced independently for atoms and electrons, and then combined.

1. Universal Half-Reaction Protocol

Step Number	Operation in Acidic Medium	Operation in Basic Medium
Step 1: Split	Split skeletal ionic reaction into OHR and RHR.	Split skeletal ionic reaction into OHR and RHR.
Step 2: Balance other atoms	Balance all atoms other than H and O first.	Balance all atoms other than H and O first.
Step 3: Balance O atoms	Add H_2O to the oxygen-deficient side.	Add H_2O to the oxygen-deficient side.
Step 4: Balance H atoms	Add H^+ to the hydrogen-deficient side.	Add H_2O to H-deficient side and equal OH^- to opposite side (or balance with H^+ first, then add OH^- to neutralize H^+).
Step 5: Balance charge	Add **electrons (e^-)** to the more positive side.	Add **electrons (e^-)** to the more positive side.
Step 6: Equalize e^- & Add	Multiply half-reactions by integers so e^- cancel, then sum and cancel duplicate H_2O/H^+.	Multiply by integers so e^- cancel, then sum and cancel duplicate species.

Table 2.1 — Stepwise rules for ion-electron balancing in acidic vs basic media.

2. Worked Exemplar: Permanganate + Iodide in Basic Medium (NCERT Problem 7.9)

KEY RESULT

Reaction: **$MnO_4^- + I^- \rightarrow MnO_2 + IO_3^-$ (Basic Medium)**

- **Oxidation Half-Reaction:** $I^- \rightarrow IO_3^-$
- Add 3 H_2O to left: $I^- + 3\ H_2O \rightarrow IO_3^-$
- Add 6 H^+ to right: $I^- + 3\ H_2O \rightarrow IO_3^- + 6\ H^+$
- Add 6 e^- to right: $I^- + 3\ H_2O \rightarrow IO_3^- + 6\ H^+ + 6\ e^-$
- **Reduction Half-Reaction:** $MnO_4^- \rightarrow MnO_2$
- Add 2 H_2O to right: $MnO_4^- \rightarrow MnO_2 + 2\ H_2O$
- Add 4 H^+ to left: $MnO_4^- + 4\ H^+ \rightarrow MnO_2 + 2\ H_2O$
- Add 3 e^- to left: $MnO_4^- + 4\ H^+ + 3\ e^- \rightarrow MnO_2 + 2\ H_2O$
- Multiply RHR by 2 to equalize electrons (6 e^-):

$2\ MnO_4^- + 8\ H^+ + 6\ e^- \rightarrow 2\ MnO_2 + 4\ H_2O$

- Add OHR and RHR: $2\ MnO_4^- + I^- + 2\ H^+ \rightarrow 2\ MnO_2 + IO_3^- + H_2O$
- Convert to Basic Medium: add 2 OH^- to both sides ($2\ H^+ + 2\ OH^- = 2\ H_2O$):

$2\ MnO_4^- + I^- + H_2O \rightarrow 2\ MnO_2 + IO_3^- + 2\ OH^-$ (Completely Balanced!)

The n-Factor (Valence Factor) — The Bridge Between Moles and Equivalents

The n-factor represents the number of moles of electrons transferred, H^+ ions replaced, OH^- ions furnished, or total cationic/anionic charge per mole of the substance in a specific reaction.

1. Universal Mathematical Definitions

KEY RESULT

- **Equivalent Weight (E): E = Molar Mass (M) / n-factor**
- **Number of Gram Equivalents (eq): eq = Weight (w) / E = moles (n) × n-factor**
- **Normality (N): N = Molarity (M) × n-factor**

2. n-factor for Non-Redox Species

Substance Category	Operational Rule for n-factor	Classic Examples & Key Traps
Acids	**Basicity:** Number of replaceable H^+ ions furnished **in that specific reaction**	· HCl: n = 1; H_2SO_4: n = 2 (in full neutralization) · H_2SO_4 + NaOH → $NaHSO_4$ + H_2O: **n = 1**! · **H_3PO_4:** Tribasic, **n = 3** (3 P—OH) · **H_3PO_3:** Dibasic, **n = 2** (2 P—OH, 1 non-ionisable P—H!) · **H_3PO_2:** Monobasic, **n = 1** (1 P—OH, 2 non-ionisable P—H!) · **H_3BO_3 (Boric acid):** Monobasic Lewis acid, **n = 1** (accepts OH^-: $B(OH)_4^-$)
Bases	**Acidity:** Number of replaceable OH^- ions furnished in that reaction	· NaOH: n = 1; $Ba(OH)_2$: n = 2 · $Al(OH)_3$ + 2 HCl → $Al(OH)Cl_2$ + 2 H_2O: **n = 2**!
Neutral Salts (No redox)	**Total positive or negative electrical charge** carried by ions per formula unit	· NaCl: n = 1; Na_2SO_4: n = 2; $AlCl_3$: n = 3 · **$Al_2(SO_4)_3$:** 2 Al^{3+} (+6) or 3 SO_4^{2-} (−6) ⇒ **n = 6** · **Potash Alum [$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$]:** K_2SO_4 (2) + $Al_2(SO_4)_3$ (6) ⇒ **n = 8**

Table 2.2 — Systematic determination of n-factor for non-redox substances.

Redox n-Factor — Net Electron Transfer per Formula Unit

In redox reactions, ***n-factor = Total change in oxidation number per molecule/formula unit of reactant =*** *Number of electrons gained or lost per mole of reactant.*

1. Acids & Bases

n-factor = Basicity / Acidity

H_3PO_4: n = 3 (tribasic)

H_3PO_3: n = 2 (1 P-H, dibasic)

H_3PO_2: n = 1 (2 P-H, monobasic)

$Al(OH)_3 + 2HCl \rightarrow Al(OH)Cl_2$: n = 2!

2. $KMnO_4$ (M.W. = 158)

Acidic: $MnO_4^- \rightarrow Mn^{2+}$ (Δ = 5)

n = 5 ⇒ E = M / 5 = 31.6

Neutral/Weak Alk: $\rightarrow MnO_2$ (Δ = 3)

n = 3 ⇒ E = M / 3 = 52.7

Strongly Alkaline: $\rightarrow MnO_4^{2-}$ (Δ = 1)

n = 1 ⇒ E = M / 1 = 158

3. Redox Salts & Disprop.

$K_2Cr_2O_7$ (Acidic): 2Cr(+6) → 2Cr(+3)

n = 2 × 3 = 6 ⇒ E = M / 6

$FeC_2O_4 \rightarrow Fe^{3+} + 2CO_2$ (both ox!)

n = 1(Fe) + 2(C) = 3 ⇒ E = M / 3

Disproportionation: 1/n = $1/n_1$ + $1/n_2$

E.g. Cl_2 in cold alkali: n = (1×1)/(1+1) = 0.5

Fig. 2.1 — Universal n-factor matrix: KMnO4 in 3 media, $K_2Cr_2O_7$, and Ferrous Oxalate.

1. Potassium Permanganate (KMnO4, M = 158 g/mol) in Three Media

Medium	Reduction Half-Reaction	Change in O.S. (Δ)	n-factor	Equivalent Weight (E)
Acidic Medium (dil H_2SO_4)	MnO_4^-(+7) + 8 H^+ + 5 e^- → **Mn^{2+}(+2)** + 4 H_2O	+7 → +2 (Δ = 5)	**n = 5**	**E = M / 5 = 31.6**
Neutral or Faintly Alkaline	MnO_4^-(+7) + 2 H_2O + 3 e^- → **MnO_2 (+4)** + 4 OH^-	+7 → +4 (Δ = 3)	**n = 3**	**E = M / 3 = 52.7**
Strongly Alkaline Medium	MnO_4^-(+7) + e^- → **MnO_4^{2-}(+6)** (Manganate ion, green)	+7 → +6 (Δ = 1)	**n = 1**	**E = M / 1 = 158.0**

Table 2.3 — Medium-dependent redox behavior of Potassium Permanganate.

2. Common Reductants and their n-Factors in JEE

Reductant Species	Oxidation Half-Reaction	Atoms Changing O.S.	n-factor Calculation	Resulting n-factor
Potassium Dichromate ($K_2Cr_2O_7$)	$Cr_2O_7^{2-}$(+6) + 14 H^+ + 6 e^- → 2 Cr^{3+}(+3)	2 Cr atoms per molecule	2 × (6 − 3)	**n = 6** (E = M/6 = 49.0)
Oxalic Acid / Oxalate ($H_2C_2O_4$ / $C_2O_4^{2-}$)	$C_2O_4^{2-}$(+3) → 2 CO_2(+4) + 2 e^-	2 C atoms per molecule	2 × (4 − 3)	**n = 2** (E = M/2)
Ferrous Sulphate ($FeSO_4$)	Fe^{2+}(+2) → Fe^{3+}(+3) + e^-	1 Fe atom	1 × (3 − 2)	**n = 1** (E = M/1)
Ferrous Oxalate (FeC_2O_4)	Fe^{2+} → Fe^{3+} + e^- AND $C_2O_4^{2-}$ → 2 CO_2 + 2 e^-	**Both cation and anion oxidised!**	1 (from Fe) + 2 (from C_2O_4)	**n = 3** (E = M/3)
Ferric Oxalate [$Fe_2(C_2O_4)_3$]	Fe is already +3 (no ox); 3 $C_2O_4^{2-}$ → 6 CO_2 + 6 e^-	6 C atoms	3 × 2 electrons lost	**n = 6** (E = M/6)
Sodium Thiosulphate ($Na_2S_2O_3$)	2 $S_2O_3^{2-}$(+2) → $S_4O_6^{2-}$(+2.5) + 2 e^- (with I_2)	2 S atoms per molecule	2 × (2.5 − 2)	**n = 1** (E = M/1)

Table 2.4 — Universal reference table for common redox partners in titrations.

Advanced n-Factor Calculations — Disproportionation and Multiple Redox Sites

When a single species contains multiple atoms undergoing oxidation/reduction in different directions, or undergoes disproportionation, calculate the effective n-factor using total electron exchange.

1. Disproportionation n-Factor Formula

KEY RESULT

For an element undergoing disproportionation where one mole undergoes oxidation with change n_1 and reduction with change n_2:

· **If 1 mole of reactant splits into both paths:**

$1 / n_{eq} = 1 / n_1 + 1 / n_2 \Rightarrow$ n-factor $= (n_1 \cdot n_2) / (n_1 + n_2)$

· **Example 1: Chlorine in Cold/Dilute Alkali:**

$Cl_2(0) \rightarrow Cl^-(-1)$ [$n_1 = 1$ e^- per Cl] AND $Cl_2(0) \rightarrow ClO^-(+1)$ [$n_2 = 1$ e^- per Cl]

n-factor of Cl_2 = **$(1 \times 1) / (1 + 1) = 0.5$** per Cl atom (or **1** per Cl_2 molecule: 2 Cl atoms transfer 1 electron total).

· **Example 2: Chlorine in Hot/Concentrated Alkali:**

$Cl_2(0) \rightarrow 5\ Cl^-(-1)$ (5 e^- gained) AND $Cl_2(0) \rightarrow ClO_3^-(+5)$ (5 e^- lost)

n-factor of Cl_2 = **$(1 \times 5) / (1 + 5) = 5/6$** per Cl atom (or **5/3** per Cl_2 molecule).

2. Multi-Element Redox (Cuprous Sulphide Cu2S)

Consider the oxidation of **Cuprous Sulphide (Cu_2S)** to **$Cu^{2+} + SO_2$** in acidic medium:

· Cu goes from **+1 to +2**: 2 Cu atoms lose **$2 \times 1 = 2$ electrons**.

· S goes from **−2 to +4**: 1 S atom loses **6 electrons**.

· **Total electrons lost per mole of $Cu_2S = 2 + 6 = 8$ electrons!**

· Therefore, **n-factor of $Cu_2S = 8 \Rightarrow$ Equivalent Weight $E = M / 8$!**

3. Partial Redox Reactions

- **When a reagent acts BOTH as a redox agent and as an acid/base (salt former):**
  Example: **$2\ KMnO_4 + 16\ HCl \rightarrow 2\ KCl + 2\ MnCl_2 + 5\ Cl_2 + 8\ H_2O$**
  - Out of 16 moles of HCl: **10 moles are oxidised** to 5 Cl_2 (lose 10 e^-), while **6 moles act merely as acid** (forming KCl and $MnCl_2$ without changing O.S. −1).
  - **n-factor of HCl in this reaction = (Total e^- lost) / (Total moles of HCl) = 10 / 16 = 5/8!**
  - Equivalent weight of HCl = **$M / (5/8) = 1.6\ M$**.

The Law of Equivalence — Stoichiometric Calculations Without Balanced Equations

The golden rule of volumetric quantitative analysis: **In any chemical reaction, 1 gram equivalent of any reactant always reacts completely with 1 gram equivalent of any other reactant to produce 1 gram equivalent of each product.**

1. Universal Equivalence Relations

KEY RESULT

Law of Chemical Equivalence: For a general reaction $a\,A + b\,B \rightarrow$ Products:

Equivalents of A = Equivalents of B = Equivalents of each Product

- In terms of mass and equivalent weight: $w_A / E_A = w_B / E_B$
- In terms of Molarity and Volume (in mL): $(M_A \cdot n_A) \cdot V_A(mL) = (M_B \cdot n_B) \cdot V_B(mL)$
- In terms of Normality and Volume: $N_A \cdot V_A = N_B \cdot V_B$
- **Number of milliequivalents (meq):** $meq = N \times V(mL) = M \times n\text{-factor} \times V(mL)$

2. Normality of Resulting Mixtures

Type of Mixture	Governing Condition	Resulting Normality Formula
Two Similar Reactants (e.g. two acids or two oxidants)	Reagents do not react with each other; equivalents are additive	$N_{final} = (N_1V_1 + N_2V_2) / (V_1 + V_2)$
Opposing Reactants (Acid + Base / Oxidant + Reductant)	Neutralization / Redox reaction consumes equivalents	If $N_1V_1 > N_2V_2$: $N_{residual} = (N_1V_1 - N_2V_2) / (V_1 + V_2)$
Exact Equivalence Point	Complete quantitative consumption with zero residual	$N_1V_1 = N_2V_2$ (Endpoint of titration)

Table 2.5 — Mixture normality equations for quantitative volumetric analysis.

3. Advantages Over the Mole Method

In the mole method, a balanced chemical equation with correct stoichiometric coefficients is strictly mandatory. In the **equivalent method**, stoichiometry is already absorbed into the n-factor! You do NOT need to balance the overall equation — only know the initial and final oxidation states.

Quantitative Redox Titrimetry (NCERT 7.3.3) — Analytical Applications

Redox titrations determine the concentration of unknown reductants or oxidants using solutions of known strength. Detection of the stoichiometric equivalence point depends on distinct visual indicator mechanisms.

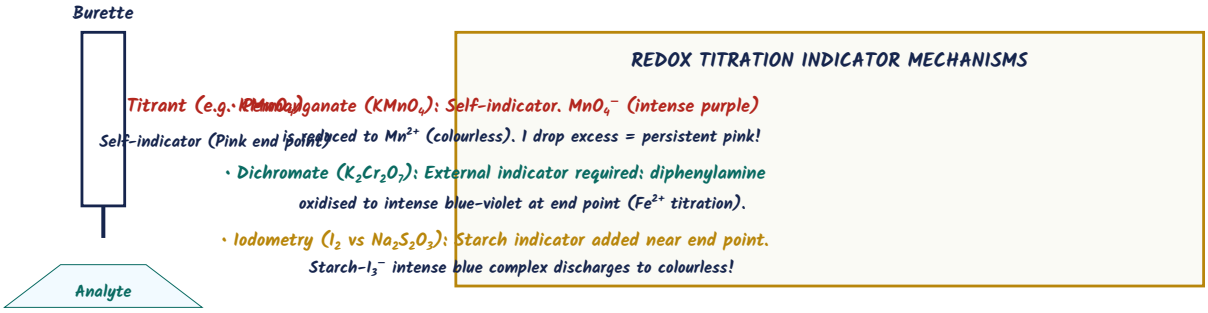


Fig. 3.1 — Redox titration apparatus and indicator mechanisms: Permanganometry, Dichromatometry, and Iodometry.

1. Permanganometry (KMnO4 Titrations)

- **Self-Indicator Action:** $KMnO_4$ acts as its own internal indicator. The intense purple **MnO_4^-** ion is reduced to nearly colourless **Mn^{2+}** in acidic solution. The very first slight excess drop of $KMnO_4$ imparts an unreacted, permanent pale pink colour to the solution.
- **Acid Medium Selection (CRITICAL JEE TRAP):**n · **Dilute H_2SO_4 is strictly used:** It does not react with $KMnO_4$ and does not undergo redox decomposition.n · **HCl CANNOT BE USED:** $KMnO_4$ is strong enough to oxidise Cl^- to Cl_2 gas (**$2\,KMnO_4 + 16\,HCl \rightarrow 2\,MnCl_2 + 5\,Cl_2\uparrow$**), consuming titrant erroneously.n · **HNO_3 CANNOT BE USED:** HNO_3 is itself a strong oxidising agent; it competes with $KMnO_4$ by oxidising the analyte.

2. Dichromatometry (K2Cr2O7 Titrations)

$K_2Cr_2O_7$ is a primary standard (obtained highly pure and stable, unlike $KMnO_4$). Reduced from orange **$Cr_2O_7^{2-}$ (+6)** to green **Cr^{3+} (+3)**.n· Because green Cr^{3+} masks visual colour changes, **an external redox indicator (Diphenylamine)** is added. At the end point, excess dichromate oxidises diphenylamine to an intense deep **blue-violet** diphenylbenzidine violet complex.n· **HCl can be used here!** Dichromate is a milder oxidant than permanganate and cannot oxidise dilute Cl^- to Cl_2.

3. Iodometry vs Iodimetry (JEE Advanced Distinction)

Feature	Iodimetry (Direct Titration)	Iodometry (Indirect Titration)
Principle	**Direct titration** of a reducing agent using a standard solution of Iodine (I_2).	**Indirect titration** of an oxidising agent by adding excess KI, then titrating liberated I_2 against standard $Na_2S_2O_3$.
Standard Solution Used	Standard **I_2 solution in KI** (forming I_3^-)	Standard **Sodium Thiosulphate ($Na_2S_2O_3$)**
Analytes Determined	Mild reducing agents: AsO_3^{3-}, Sn^{2+}, H_2S, SO_3^{2-}	Strong oxidising agents: Cu^{2+}, $Cr_2O_7^{2-}$, ClO^-, MnO_4^-, H_2O_2, O_3
The Classic Cu^{2+} Reaction	—	**$2\,Cu^{2+} + 4\,I^- \rightarrow Cu_2I_2(s)\downarrow$ (white ppt) + I_2** Followed by: **$I_2 + 2\,S_2O_3^{2-} \rightarrow 2\,I^- + S_4O_6^{2-}$**
Indicator Timing Trap	Starch added at beginning.	**Starch MUST BE ADDED NEAR THE END POINT** (when solution turns pale straw yellow)! If added too early, I_2 is trapped irreversibly inside the starch helix.

Table 3.1 — Fundamental analytical contrast between Iodimetry and Iodometry.

Electrochemical Cells (NCERT 7.4) — Converting Chemical Energy to Electrical Energy

When a redox reaction is carried out indirectly in two separate compartments, electron transfer cannot occur directly. Electrons are forced through an external metallic circuit, generating electric current.

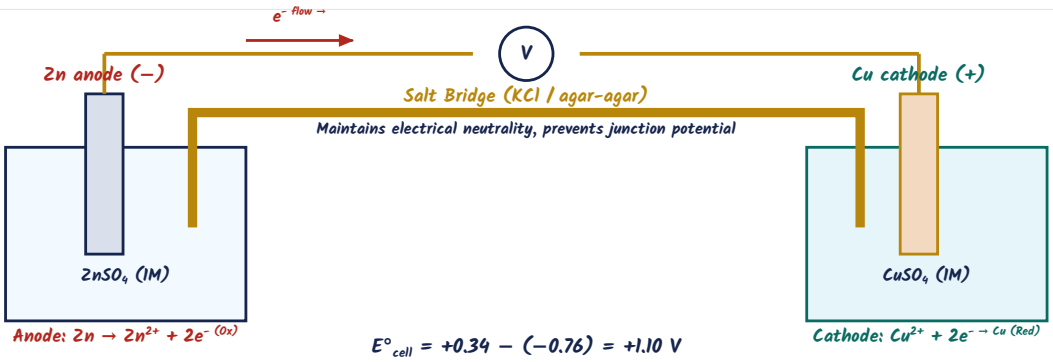


Fig. 3.2 — The Daniell galvanic cell showing anode oxidation, cathode reduction, salt bridge, and electron flow.

1. Architecture of the Daniell Cell

Component	Anode Half-Cell (Left)	Cathode Half-Cell (Right)
Electrode & Solution	**Zinc rod** immersed in 1.0 M $ZnSO_4$ solution	**Copper rod** immersed in 1.0 M $CuSO_4$ solution
Half-Reaction Occurring	**Oxidation:** $Zn(s) \rightarrow Zn^{2+}(aq) + 2\ e^-$	**Reduction:** $Cu^{2+}(aq) + 2\ e^- \rightarrow Cu(s)$
Electrical Polarity	**Negative (−)** (Source of electrons)	**Positive (+)** (Sink for electrons)
Electrode Mass Change	Zinc rod **dissolves / loses mass**	Copper rod **deposits / gains mass**
Standard Potential ($E°$)	$E°(Zn^{2+}/Zn)$ = **−0.76 V**	$E°(Cu^{2+}/Cu)$ = **+0.34 V**
Cell EMF ($E°_{cell}$)	$E°_{cell} = E°_{cathode} - E°_{anode}$ = +0.34 V − (−0.76 V) = **+1.10 V**	

Table 3.2 — Comprehensive operational breakdown of the Daniell cell.

2. The Functions of the Salt Bridge

KEY RESULT

The salt bridge is an inverted U-tube filled with an inert electrolyte (KCl, KNO_3, or NH_4NO_3) in agar-agar gel.n· **Function 1 (Completes the Circuit):** Allows migration of ions between the two half-cells, maintaining electrical continuity.n· **Function 2 (Maintains Electrical Neutrality):** As Zn^{2+} accumulates at anode, Cl^- migrates into anode beaker; as Cu^{2+} is consumed at cathode, K^+ migrates into cathode beaker.n· **Function 3 (Eliminates Liquid Junction Potential):** Prevents the accumulation of opposing charges at the liquid-liquid boundary.n· **Inertness Constraint:** Velocities (transference numbers) of cation and anion must be nearly identical ($u_+ \approx u_-$). **KCl cannot be used if half-cell contains Ag^+, Pb^{2+}, or Hg_2^{2+}** because insoluble precipitates (AgCl, $PbCl_2$) form and clog the bridge!

Electrochemical Series (NCERT Table 7.1) — The Absolute Measure of Redox Power

Electrode potential (E) is the potential difference between the metal electrode and its surrounding ions in solution. By universal IUPAC convention, **all standard potentials (E°) are tabulated as Standard Reduction Potentials (SRP)** at 298 K, 1 atm, and 1.0 M concentration.

1. Standard Hydrogen Electrode (SHE / NHE) – The Universal Datum

The potential of an individual isolated half-cell cannot be measured. The **Standard Hydrogen Electrode (SHE)** is assigned a potential of **identically 0.00 V at all temperatures**:
$2\ H^+(aq, 1.0\ M) + 2\ e^- \rightleftharpoons H_2(g, 1\ bar)$ $(E° = 0.00\ V)$
· Platinized platinum foil coated with finely divided platinum black adsorbs H_2 gas and establishes rapid thermodynamic equilibrium with aqueous H^+.

2. Master Trends in the Electrochemical Series

Redox Couple (Ox / Red)	Standard Reduction Potential (E°)	Chemical Implications & Behavior
$F_2(g)$ / F^-	**+2.87 V (Highest!)**	**Strongest oxidising agent** known. Fluorine has the greatest tendency to gain electrons and be reduced to F^-. Cannot be oxidised by any chemical reagent.
$S_2O_8^{2-}$ / SO_4^{2-}	**+2.01 V**	Very powerful peroxo oxidant.
MnO_4^- / Mn^{2+} (acidic)	**+1.51 V**	Strong oxidising agent in acidic titrations.
$Cl_2(g)$ / Cl^-	**+1.36 V**	Strong oxidant; oxidises Br^- and I^-.
$Cr_2O_7^{2-}$ / Cr^{3+} (acidic)	**+1.33 V**	Primary standard oxidant.
$O_2(g)$ / H_2O	**+1.23 V**	Thermodynamic boundary for water oxidation.
$Br_2(l)$ / Br^-	**+1.09 V**	Moderate oxidant.
Ag^+ / Ag	**+0.80 V**	Easily reduced to metal.
Fe^{3+} / Fe^{2+}	**+0.77 V**	Reversible biochemical / titrimetric couple.
$I_2(s)$ / I^-	**+0.54 V**	Mild oxidant / I^- is moderate reductant.
Cu^{2+} / Cu	**+0.34 V**	Positive SRP: unreactive towards non-oxidising acids (does not liberate H_2 from HCl).
$2\ H^+$ / $H_2(g)$	**0.00 V (Arbitrary Datum)**	**Reference benchmark** separating active from noble metals.
Fe^{2+} / Fe	**−0.44 V**	Active metal: dissolves in acid with H_2 liberation.
Zn^{2+} / Zn	**−0.76 V**	Standard anode; reducing agent.
Al^{3+} / Al	**−1.66 V**	Powerful reductant (thermite process).
Mg^{2+} / Mg	**−2.36 V**	Reacts with steam / hot water.
Na^+ / Na	**−2.71 V**	Violently reacts with cold water.
Li^+ / Li	**−3.05 V (Lowest!)**	**Strongest reducing agent in aqueous solution!** Lithium has the lowest SRP due to immense **Hydration Enthalpy**.

Table 7.1 — Standard reduction potentials of selected couples at 298 K.

Thermodynamic Predictions from E° — Feasibility Criteria and Latimer Diagrams

Standard reduction potentials provide the exact quantitative basis to predict whether a chemical reaction will occur spontaneously, whether a metal will liberate hydrogen, and whether a species will disproportionate.

1. Spontaneity Rule of Redox Reactions

KEY RESULT

- Standard Cell Potential: $\mathbf{E^\circ_{cell} = E^\circ_{cathode}}$ **(SRP of reduction)** $\mathbf{- E^\circ_{anode}}$ **(SRP of oxidation)**
- **Thermodynamic Criterion:** A redox reaction is spontaneous under standard conditions if and only if: $\mathbf{E^\circ_{cell} > 0}$ **(Positive EMF)** $\Leftrightarrow \mathbf{\Delta G^\circ = -n\,F\,E^\circ_{cell} < 0}$
- **Relative Position Rule:** An oxidising agent can oxidise any reducing agent that lies **BELOW it in the reduction potential series** (more negative SRP).

2. Liberation of Hydrogen Gas from Dilute Acids

For a metal to displace H_2 from dilute acids ($2\,H^+ + 2\,e^- \rightarrow H_2$, $E^\circ = 0.00$ V):

- **Requirement:** Metal must have a **NEGATIVE SRP** ($E^\circ < 0.00$ V). E.g. Zn (−0.76 V), Fe (−0.44 V), Mg (−2.36 V).
- Metals with **POSITIVE SRP** (Cu +0.34 V, Ag +0.80 V, Au +1.50 V) **CANNOT liberate H_2 from dilute HCl or dil H_2SO_4!** (They dissolve only in oxidising acids like concentrated HNO_3 through reduction of nitrogen, not reduction of H^+!).

3. Latimer Diagrams & The Disproportionation Test (JEE Advanced)

A **Latimer diagram** writes oxidation states of an element from left to right in decreasing order:
$A \xrightarrow{(E_1^\circ)} B \xrightarrow{(E_2^\circ)} C$

- **Disproportionation Criterion:** Species B will **spontaneously disproportionate** into A and C if and only if: $\mathbf{E^\circ_{right}}$ **(E°_2)** $\mathbf{> E^\circ_{left}}$ **(E°_1)**
- **Rationale:** In disproportionation: B is reduced to C (Potential = $+E^\circ_2$) and B is oxidised to A (Potential = $-E^\circ_1$). Net EMF = $\mathbf{E^\circ_2 - E^\circ_1 > 0}$!
- **Example (Copper in aqueous solution):**

$Cu^{2+} \xrightarrow{(+0.15\ V)} Cu^+ \xrightarrow{(+0.52\ V)} Cu$

Here, E°_{right} (+0.52 V) > E°_{left} (+0.15 V).

Therefore, **Cu^+ spontaneously disproportionates in water: $2\,Cu^+ \rightarrow Cu^{2+} + Cu(s)$** ($E^\circ_{net} = +0.37$ V)!

Thermodynamic Interrelations — Connecting EMF, Free Energy & Equilibrium

The electrical work performed by a galvanic cell is quantitatively linked to the decrease in Gibbs free energy and the thermodynamic equilibrium constant of the chemical reaction.

1. Gibbs Free Energy and Cell Potential

KEY RESULT

Fundamental Thermodynamic Equation: $n\Delta G^\circ = -n \cdot F \cdot E^\circ_{cell}$

where:
- **n** = number of moles of electrons transferred in the balanced redox equation.
- **F** = Faraday's constant ≈ **96,485 C/mol** (charge carried by 1 mole of electrons).
- **E°_{cell}** = standard electromotive force of the cell in Volts (J/C).
- **Important Property:** ΔG° is an **extensive property** (additive), whereas E° is an **intensive property** (NOT directly additive!).

2. Calculation of Non-Adjacent Half-Cell Potentials (Critical Trap)

To find the potential for a composite couple from two consecutive steps ($A \xrightarrow{(E_1^\circ)} B \xrightarrow{(E_2^\circ)} C$, net $A \xrightarrow{(E_3^\circ)} C$):

- **NEVER add potentials directly:** $E_3^\circ \neq E_1^\circ + E_2^\circ$!
- **MUST add Gibbs energies:** $\Delta G_3^\circ = \Delta G_1^\circ + \Delta G_2^\circ$

$-n_3 F E_3^\circ = -n_1 F E_1^\circ + (-n_2 F E_2^\circ)$

$E_3^\circ = (n_1 E_1^\circ + n_2 E_2^\circ) / n_3$

- **Classic Problem:** Given $Fe^{3+} + e^- \rightarrow Fe^{2+}$ ($E_1^\circ = +0.77$ V) and $Fe^{2+} + 2e^- \rightarrow Fe$ ($E_2^\circ = -0.44$ V):

For $Fe^{3+} + 3e^- \rightarrow Fe$: $E_3^\circ = (1 \times 0.77 + 2 \times (-0.44)) / 3 = -0.037$ V!

3. Relation with Equilibrium Constant (K)

At 298 K, substituting R = 8.314 J/(mol K), T = 298 K, and F = 96,485 C/mol:

$\Delta G^\circ = -RT \ln K = -2.303\ RT \log_{10} K$

$E^\circ_{cell} = (2.303\ RT) / (n F) \log_{10} K = (0.0591 / n) \log_{10} K$

- For the Daniell cell ($n = 2$, $E^\circ = 1.10$ V): $\log_{10} K = (2 \times 1.10) / 0.0591 = 37.22 \Rightarrow K \approx 1.7 \times 10^{37}$ (reaction goes to absolute completion!).

Master Quick-Revision Matrix — Core Principles, Decision Algorithms & Traps

A consolidated high-density review summarizing the essential rules, calculations, and analytical traps in Redox Reactions & Equivalent Concept.

1. Universal 5-Step Redox Diagnostic Protocol

- **Step 1: Assign Structural O.S.:** Check for peroxides (CrO_5 = +6, H_2SO_5 = +6, $H_2S_2O_8$ = +6), mixed oxides (Fe_3O_4, Pb_3O_4 = +8/3), or fractional chains (C_3O_2, $S_4O_6^{2-}$).
- **Step 2: Identify Redox Roles:** Max O.S. can only be oxidant; Min O.S. can only be reductant; Intermediate O.S. can be both or disproportionate.
- **Step 3: Calculate n-factor:**
  - For simple redox: n = Total change in O.S. per molecule.
  - For multiple oxidised atoms (e.g. FeC_2O_4): sum changes (1 + 2 = 3).
  - For disproportionation: $n = (n_1 \cdot n_2) / (n_1 + n_2)$.
- **Step 4: Apply Equivalence:** Equivalents of Oxidant = Equivalents of Reductant ($N_1V_1 = N_2V_2$). Avoid writing balanced equations unless asked.
- **Step 5: Check EMF & Spontaneity:** $E°_{cell} = E°_{red,cathode} - E°_{red,anode} > 0$. For Latimer diagrams, disproportionation requires $E°_{right} > E°_{left}$.

2. Master Summary Table of Key Reagents in Redox Titrations

Reagent	Medium / Reaction Conditions	n-factor	Equivalent Weight (E)	Special Feature / Indicator
$KMnO_4$	Acidic (dil H_2SO_4)	**5**	**M / 5 = 31.6**	Self-indicator (pink endpoint). HCl cannot be used.
$KMnO_4$	Neutral / Faintly Alkaline	**3**	**M / 3 = 52.7**	Forms brown MnO_2 precipitate.
$KMnO_4$	Strongly Alkaline	**1**	**M / 1 = 158.0**	Forms green manganate $[MnO_4]^{2-}$.
$K_2Cr_2O_7$	Acidic (H_2SO_4 or HCl)	**6**	**M / 6 = 49.0**	Primary standard; external indicator: diphenylamine.
$H_2C_2O_4 \cdot 2H_2O$	Acidic (titration with $KMnO_4$)	**2**	**M / 2 = 63.0**	Primary standard oxalic acid; titration heated to ≈60°C.
$Na_2S_2O_3 \cdot 5H_2O$	Iodometry (with I_2)	**1**	**M / 1 = 248.2**	Starch indicator added near straw-yellow endpoint.
$FeSO_4 \cdot 7H_2O$	Redox titrations	**1**	**M / 1 = 278.0**	Green vitriol; oxidised $Fe^{2+} \rightarrow Fe^{3+}$.
FeC_2O_4	Acidic (with $KMnO_4$)	**3**	**M / 3**	Both Fe^{2+} and $C_2O_4^{2-}$ oxidised simultaneously!

Table 3.3 — Universal operational constants for redox volumetric titrations.